Lab: Add Rigging to a 3D Model

So far we've learned how to create 3D models with Blender, and how to do basic animation using Blender. Let's combine the two skills and learn how to animate a 3D modeled character. In order to do this we need to learn how to add an armature to our model. The process of adding virtual bones and controllers to affect our model is known in 3D as rigging.

Overview



Open *Lab-Rigging.blend* into Blender. You will see a low polygon model of a chimpazee.

Make sure the Blender Timeline is at the bottom of your layout. If not, add a panel to the bottom of your Blender layout and in the *Editor Type* choose *Timeline*.

You can view the final animation below:

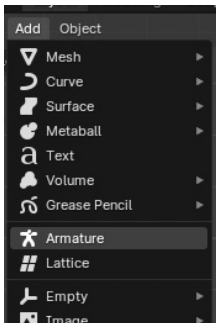
https://youtu.be/3uz6zGvecKo

Armature

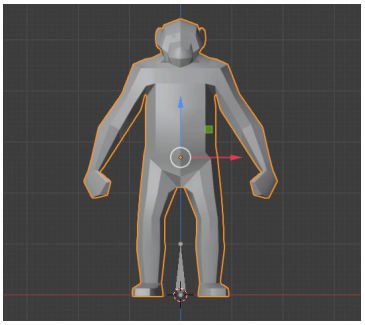
What is an armature? Imagine you have a favorite action figure or doll that you want to pose so that you can take a photo of it. In order for that to happen you would need "bones" or a framework that the plastic or cotton could adhere to, that is what the armature is, a framework. Once the armature was in place you could pose your doll or action figure.

Currently we have a 3D mesh of a chimpazee, but no way to easily move the arms and legs. If we wanted to do that we would need to move verticies and polygon faces, which would not be easy. Instead we are going to add a virtual armature to the mesh, this armature will allow for virtual bones that we can manipulate and pose.

Add an Armature

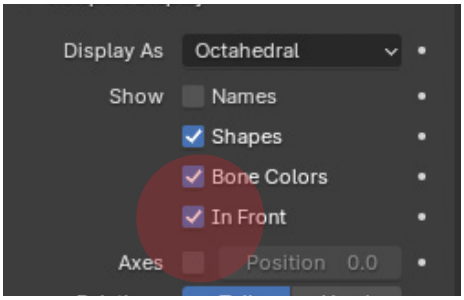


Let's start adding an armature to our chimpazee character. Put the scene into *Front* orthographic mode. To add an armature to our scene go to *Add > Armature*, the result will be a pointy arrow added to the center of our scene, this is our armature or bone system. Select the armature and use the Move Tool to place the armature at the lower pelvis area of the chimpazee.



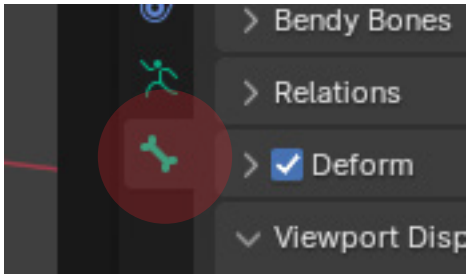
As you move the armature to the base of the chimpazee you may notice it disappears. This is because it is inside of the model itself. This makes it difficult to work with, so there are a few options. First option is you could put the world into *Wireframe Mode*, this would

make it easy to see the armature, the other option is we can tell the armature to always be in front of the model. In the Properties panel click on the *Data* tab. Under *Viewport Display* check the box marked *"In Front"*, the result will be having the armature system appear in front of the mesh, even though physically it is inside of it.

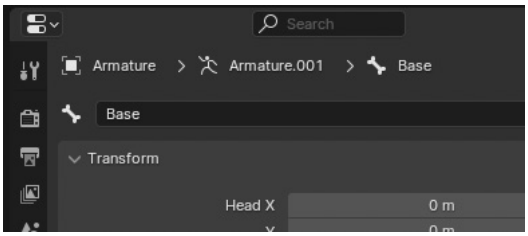


Name and Extrude Bones

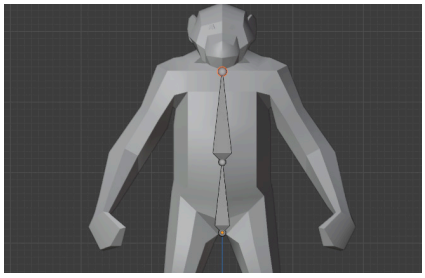
The armature is a collection of bones, so we will eventually add additional bones to our skeleton. Before we do this it is helpful to name bones, this will make animating the bones easier in the future.



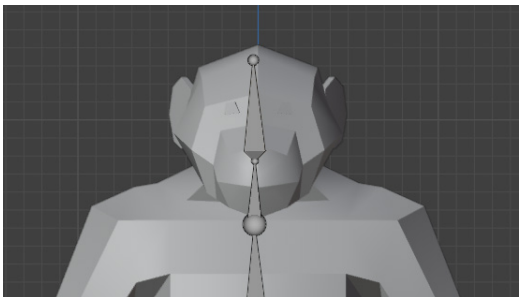
Select the bone we just positioned. In the *Properties* panel choose the *Bone* tab. At the top you can name your bones individually. If you do not give them a name they will default to a generate name like *"Bone"*. In this case let's give a descriptive term for the bone. In the tab rename it to *"Base"*.



Let's add a second bone to our armature. Select the *Base* bone and go into *Edit Mode*. In *Edit Mode* make sure the top of the bone is highlighted and *Extrude* tool to add a new bone. A keyboard shortcut is using the *"e"* key. Drag the next bone to just below the neck of the chimpanzee. Notice that the new bone is called *Base.01* and in the *Outliner* it is a subset of the *Base* bone. This is because the new bone is considered a child of the original *Base* bone. With armature a parent bone will control the child bone, but not vice versa. In other words moving the *Base* bone will move the *Base.01* bone, but not vice-versa. Rename the new bone *Torso*.

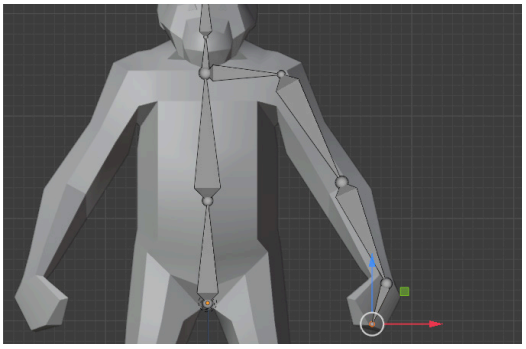


Add Arm and Leg Bones



Now that we know how to add bones, let's add more bones to our armature. Click on *Torso* and extrude a bone into the neck. Then extrude a bone to the top of the head, you can use the image on the right as a reference.

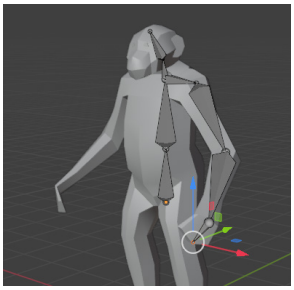
Now that we have bones in the head and neck let's add bones to the torso that go down the arm of the chimpanzee. Select the Torso bone again and extrude to the right. Create a bone for the shoulder, the bicep, the forearm and hand. Use the image on the right as a reference.



Now that we have the head / neck and right arm set for our armature let's set the right leg. Select the *Base* bone and extrude to the right. Then extrude a bone for the upper thigh, lower leg or shin.

Now that you have a basic armature / skeleton setup set for the front, we need to make sure that the bones are in the correct

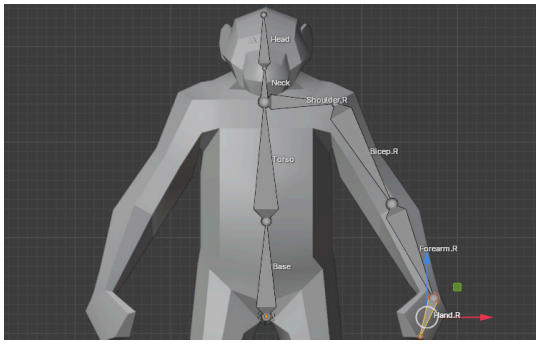
spots. Put the chimphanzee model into the *Right* orthographic view. In *Edit Mode* move the bones to fit properly into the character, make sure the bones are in the center of the model, this may mean switching orthographic views.



Fro a side view select the lower leg bone and extrude it to give you a foot.

Rename the Bones

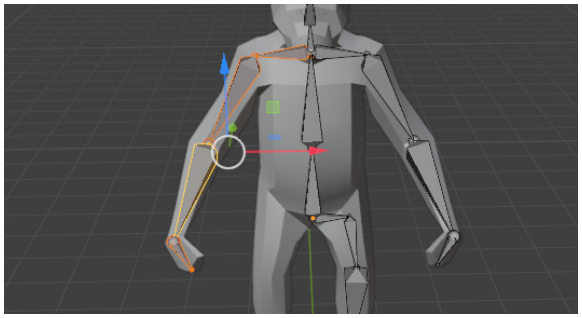
Now that we have a basic armature it is very helpful to name the bones. As we stated before naming bones can be helpful with animating. Select each bone and rename it in the *Bone* tab of the *Properties* panel.



Note that with skeleton / armatures there are times where bone structures repeat on the opposite side. For instance, the arm bones on the right would be the same bone strucutre on the left hand side, the same goes for the right leg to the left leg. When naming your bones name them with a *".r"* such as:

bicep.r
forearm.r
hand.r
etc

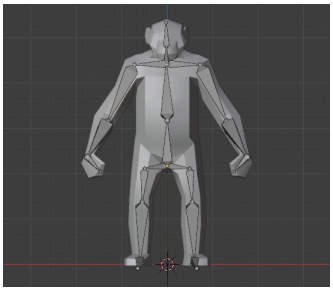
This will be use later when we copy and paste the bones.



Symmetrize Bones

Now that we have the right side of the skeleton done let's copy and paste new bones for the left hand side.

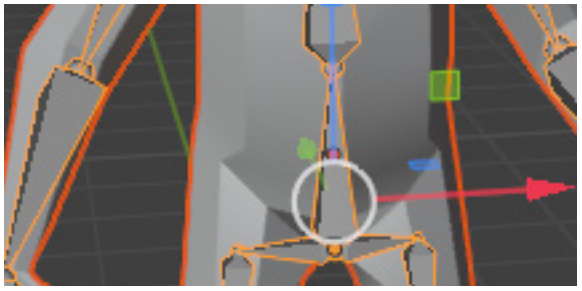
In *Edit Mode* select the shoudler bone, bicep bone, forearm bone and hand bone and go to *Armature > Symmetrize.* If you named your arm bones ending in *.r* it will repeat them on the left side with the same name, now ending in *.l.* Select the pelvis, thigh, shin and foot bone on the right side and repeat the process of symmetrize.



Your armature should look roughly like the image on the right.

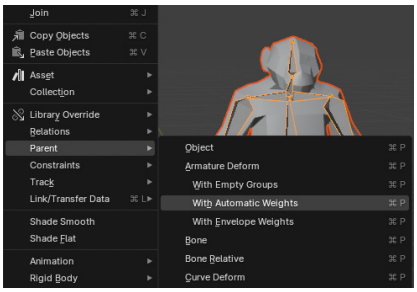
Bind Your Armature to the Model

Now that we have an armature it is time to combine that armature with the model so it can affect it. This process in 3D modeling is typically referred to as binding.



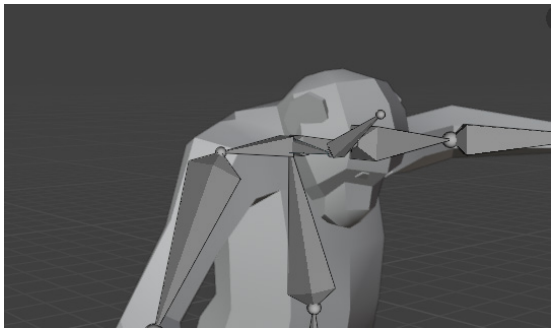
Make sure you are in *Object Mode* and select the *Chimpanzee* mesh. While holding down the *Shift* key on your keyboard select the armature so that both are highlighted. After selecting both

go to *Armature > Parent > With Automatic Weights*. The results of this is the *Chimpanzee* mesh with become a child of the *Armature* in your *Outliner*, or in other words, the armature controls the *Chimpanzee* mesh.,



Pose Your Model

Now the moment of truth, let's see if our armature is working correctly. Highlight the model and in the setting set it to *Pose Mode*. *Pose Mode*, like it sounds like, allows you to pose your model.

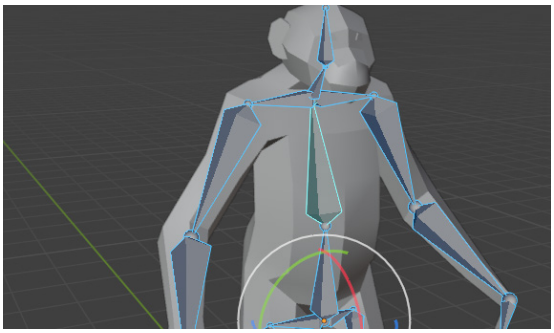


Once your model is in *Pose Mod*e highlight the right bicep bone and rotate along the y-axis. The result will be that the arm rotates upward.

Try selecting other bones and use the *Rotate* tool to adjust the pose of the character.

Reset Position

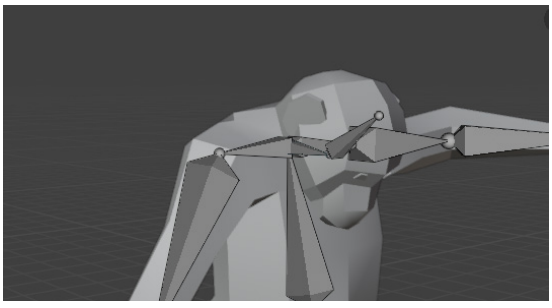
In *Pose Mode* select the *Armature* tab in the *Properties* panel. At the top you will see two buttons: *Pose Position* and *Reset Position*. The *Pose Position* is the current position of the armature, the *Reset Position* is the original position the armature was when it was attached to the 3D mesh.



If you want to reset your armature to the default pose, you can. In *Pose Mode* go to *Select > All*, this will select all of the bones in your armature automatically. Then press *Alt / Option + R*, to reset the pose back to the default.

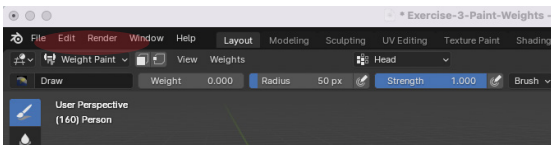
Weight Painting

When you apply an armature to a 3D mesh what occurs is each bone begins to affects certain vertices in your model. Sometimes we may want a bone to affect more or less of our model, this process is known as weight painting. Weight painting allows you to paint or control how much influence a bone has on a part of a model.



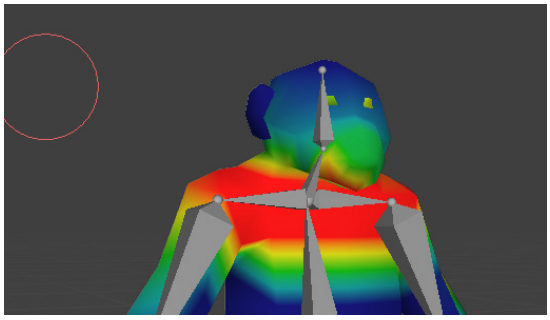
In *Pose Mode* select the *Neck* bone and rotate the neck forward along the x-axis. The result should look roughly like the image on the right. Notice how much of the model is affected.

Select the *Chimpanzee* mesh, not the armature in your scene. In the settings dropdown menu choose *Weight Painting*. The result is the model will look blue with areas of red or orange.

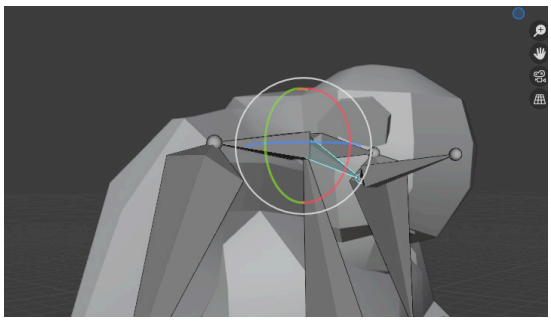


The first thing to do is select the bone we want to affect. In the dropdown menu in the top middle find the *Neck* bone and select it, the result is the neck area turn blue-ish green. With weight

painting the idea is the warmer the color, such as red or orange the stronger the influence. The cooler the color, like dark blue the less or non-existant influence the bone has on that area of the model.



Under the *Weight* option the value goes from *0* to *1*. "*1"* means maximum influence or red, *"0"* means minimum influence or "blue". Set the *Weight* to 1 and begin to click and paint on the *Chimpanzee* along the torso below the neck, apply this to the back as well, use the image on the right as a reference.

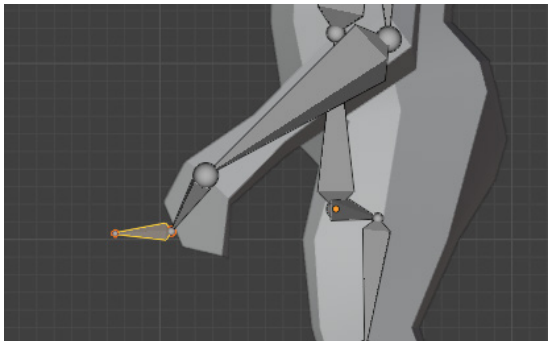


Go back into *Pose Mode* and rotate the *Neck* bone along the x-axis, notice how more of the model is affected then previously, this is due to the newly increased weights.

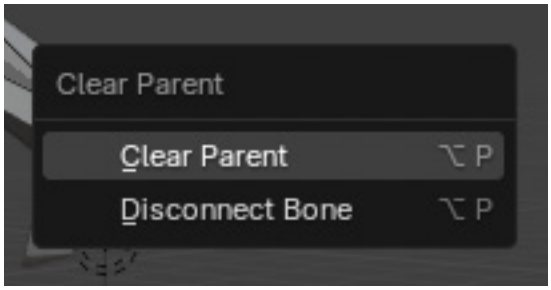
Feel free to go through the bones and figure out what you believe the appropiate weights should be.

Inverse Kinematics

Okay, so we have an armature or skeleton for our model, and we've determined how to change the influence of each bone using weight painting. One issue with posing our *Chimpanzee* character with our armature is that you always have to start top --> down. In other words, if we wanted to move the right hand we would have to start with the bicep, then move the forearm, and then move the hand.



The reason for this is forward kinematics. In forward kinematics you move the parent bone, which in turn moves the child bone, then you move down to the child and adjust it which affects it's children bones and so on. *Inverse Kinematics* is the idea that we could move the hand or foot and affect the rest of the arm or leg rather than having to start at the top, let's learn how to do this by applying inverse kinematics to the right arm.



Begin by creating what is a referred to as a controller node, this controller is what will move the entire right arm. One of the easiest ways to do this is to take a bone that is part of armature and detach it from the main armature so it acts as a parent.

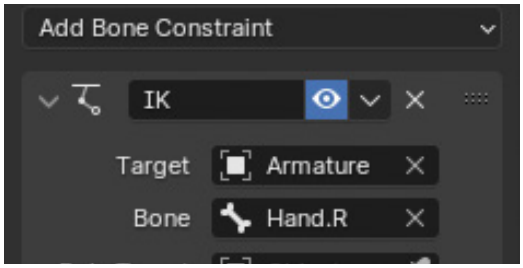
Put the armature in *Edit Mode* and select the tip of the right hand. Use the extrude tool to extrude a bone from the tip. Select the new bone and call it *Right_Arm_Controller*. Right click on the new bone and go to *Parent > Clear > Clear Parent*, this will detach the bone from being a child of the hand, but still keep it as part of the armature.



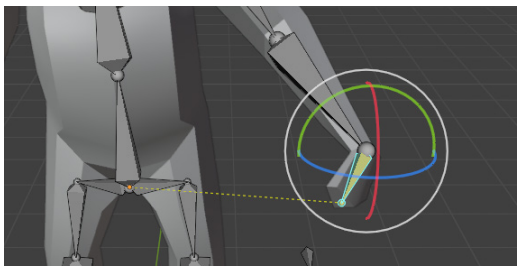
Next put the armature in *Pose Mode*. Select the *Hand.r* bone in your armature since it is the hand that we want to control the arm.

In the *Properties* panel go to *Bone Constraints*. Add a *Bone*

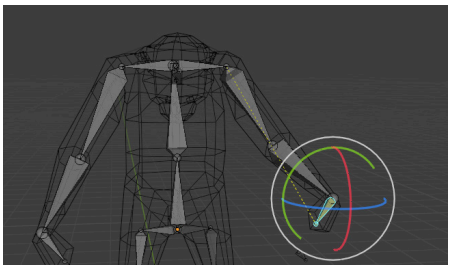
Constaraint and choose *Inverse Kinematics*. This will open a bone constrainst called IK. Under *Target* we need to say the name of the armature we are applying this to, which in this case is the *Armature* we built for this lab. Next we have to state which bone is controlling the inverse kinematics, which in this case is the new *Right_Arm_Controller.*



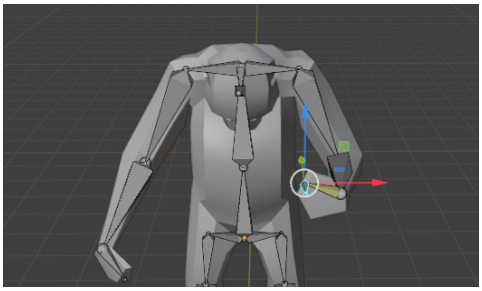
When you attach the inverse kinematics to the *Right_Arm_Controller* you'll notice that the *Chimpanzee* model tilts to the right, this is because by default the inverse kinematics goes from the bone to the original or base of the armature. You can actually see this with the yellow dotted line connecting the *Right_Arm_Controller* bone to the *Base* bone.



We don't want the inverse kinematics connecting that far up the chain, instead we want it to stop at the *Shoulder.r* bone. To fix this go back to the *IK Bone Constraint* and change *Chain Length* from *0*, which means the entire armature, to *3*. The *3* means from the starting point, which is the *Hand.R* bone, how many bones up will be affected. To see the change put the scene in *Wireframe Mode*, you can see that the yellow dotted line now connects to the shoulder blade, instead of the pelvis.



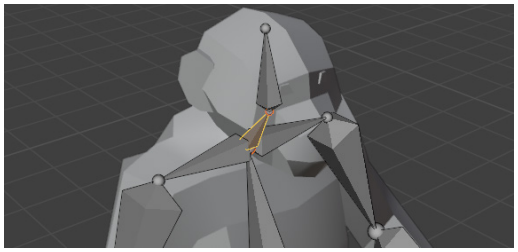
Now comes the true test, let's see if our inverse kinematics controller works. Select the *Right_Arm_Controller* in *Pose Mode* and move it left an right. When you do you will see the entire right arm moves.



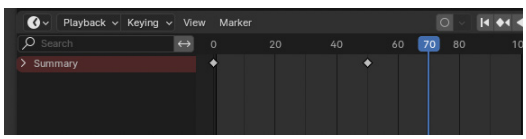
Animate a Rigged Character

Now that we have an armature / rig for our character let's animate it. Make sure that the Timeline is available in your scene.

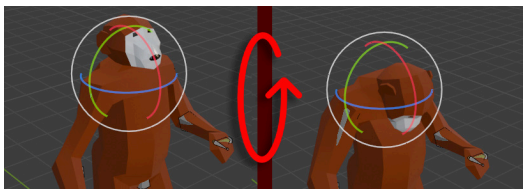
Move the blue time marker to the beginning of the *Timeline* or *frame 1*. In *Pose Mode* select the *Neck* bone and press the letter **'K'** to set a keyframe. From the popup menu choose *Location, Rotation and Scale*.



Move the *Timemarker* to *Frame 50*. Select the *Neck* bone again and rotate the bone forward along the x-axis. With the bone selected press *"k"* and from the popup menu choose *Location, Rotation and Scale*. You should see two keyframe in your timeline. Scrub through the timeline to see the *Chimpanzee* rotate it's head.

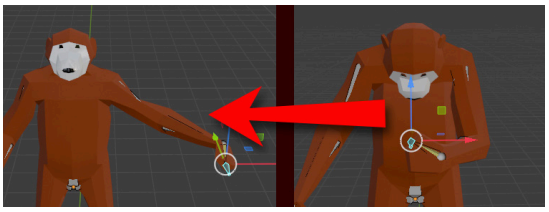


Move the blue time marker to the beginning of the *Timeline* or *frame 1*. In *Pose Mode* select the *Right_Arm_Controller* bone and press the letter *"K"* to set a keyframe. From the popup menu choose *Location, Rotation and Scale*.



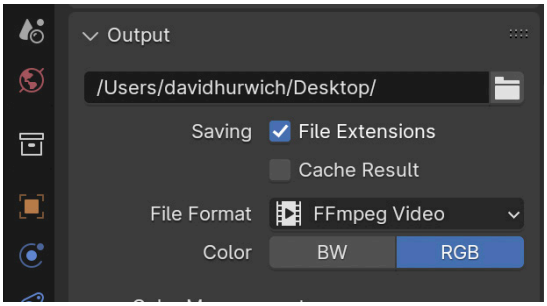
Move the *Timemarker* to *Frame 50*. Select the *Right_Arm_Controller* bone again and move the bone towards the chimpanzee along

the x-axis. With the bone selected press **“k”** and from the popup menu choose *Location, Rotation and Scale.* Scrub through the timeline to see the Chimpanzee’s arm move.



Camera and Rendering

Now that we’ve animated our character let’s export this animation. Select the camera in the scene and position it how you want it to capture the *Chimpanzee* animation. Feel free to animate the camera if you want to.



Before we render our scene, let’s double check we have the settings we want. Click on the *Output* tab in the *Properties Panel*. Double check the frame rate and resolution of the animation. You should export the animation at *24fps* and at HD format which would be *1920 x 1080px*.

Next set where the animation will be output to, this can be done in the *Output* section. I recommend setting it to the *Desktop* of your computer.

Finally set the color mode and *File Format*. Set the *Color Mode* of the video to *RGB* and set the *File Format* to *FFmpeg* Video. Under *Encoding* change it from *Matroska* to *MPEG-4*. Under View make sure the Video Codec *H.264* is being used.

Go to *Render > Render Animation* and render out your final animation.

Submission

After exporting your space animation upload it to a video sharing service such as YouTube or Vimeo. After uploading copy the url to the video and paste it into a text document. Save the document as *Rigging-Lab-URL*.

Submit your *Lab-Rigging.blend* file and *Rigging-Lab-URL* document in a zipped compressed folder to Canvas for grading.